

CS1CA Computer Systems Architecture

Information Representation – 2

Week 1 lecture 3 of the autumn semester 2025

Lectured by

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Outline of the topic

- Introduction: what is information?
- Decimal, binary and hexadecimal number systems
 - Conversion between number systems
- Internal representation of characters
- Internal representation of integers
- Arithmetic and logic operations
- Internal representation of real numbers
- IEEE 754 floating point format

In this lecture

- The following contents will be delivered.
 - Conversion between number systems
 - Internal representation of characters
- Learning outcomes
 - To be able to conduct the conversions between the number systems of binary, decimal, hexadecimal, (and octal)
 - To gain knowledge of ASCII and Unicode
 - To be able to translate a plain text to ASCII code and vice versa

Conversion between number systems – 1

- From binary (base $R = 2$) to decimal (based 10)

- Simple, just evaluate polynomial representation.

$$11010_2 = 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 26_{10}$$

- From hexadecimal (base $R = 16$) to decimal

- Simple, evaluate polynomial representation;

- There are 6 characters in hexadecimal numbers;

- The letters should be replaced by the equivalent decimal values in conversion.

$$2BC_{16} = 2 \times 16^2 + 11 \times 16^1 + 12 \times 16^0 = 700_{10}$$

Conversion between number systems – 2

- From decimal to binary (base 2)
 - Consider $N_{10} \div 2$
 - Then $N_{10} \text{ div } 2 = q_{10}^0$ and $N_{10} \text{ mod } 2 = r_0$
 - We have $N_{10} = q_{10}^0 \times 2 + r_0$
 - q_{10}^0 is divided by 2 yielding the next digit r_1 , and q_{10}^1 , and so on.
 - **Until** $q_{10}^n = 0$, and we have r_n
 - The binary number is written as $\langle r_n, \dots, r_1 r_0 \rangle_2$

Conversion between number systems – 3

- From decimal to hexadecimal (based 16)

- Consider $N_{10} \div 16$

- Then $N_{10} \text{ div } 16 = q_{10}^0$ and $N_{10} \text{ mod } 16 = r_0$

- We have $N_{10} = q_{10}^0 \times 16 + r_0$

- q_{10}^0 is divided by 16, yielding the next digit r_1 and q_{10}^1 , and so on.

- $q_{10}^n = 0$
 - Until , and we have r_n

- The hexadecimal number is $\langle r_n \dots r_1 r_0 \rangle_{16}$

Conversion between number systems – 4

- An algorithm for converting a decimal number N to a base R number.

initialise r ;

read N ;

WHILE $N > 0$ DO

$r_i := N \text{ MOD } R$; //generate the remainder

record r_i ; // record the remainder

$N := N \text{ DIV } R$; // update N by the

quotient

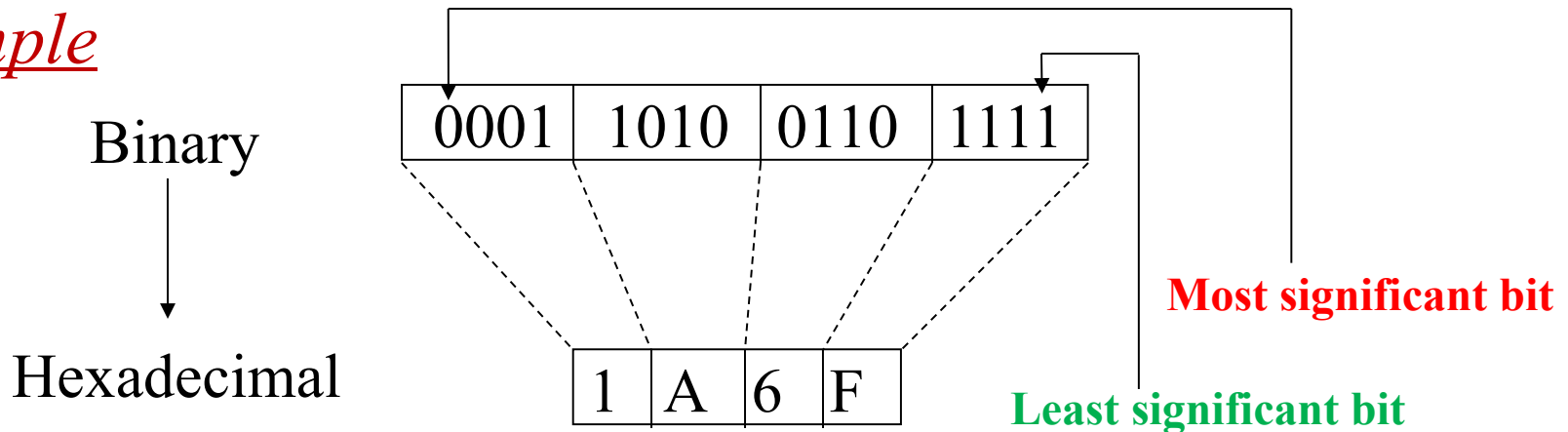
END;

The base R number: $\langle r_n, \dots, r_1, r_0 \rangle_R$

Conversion between number systems – 5

- From binary to hexadecimal
 - This is easy because $16 = 2^4$;
 - Group binary into groups of four bits from **the least significant bit**;
 - Convert each 4-bit group to decimal and then hex.

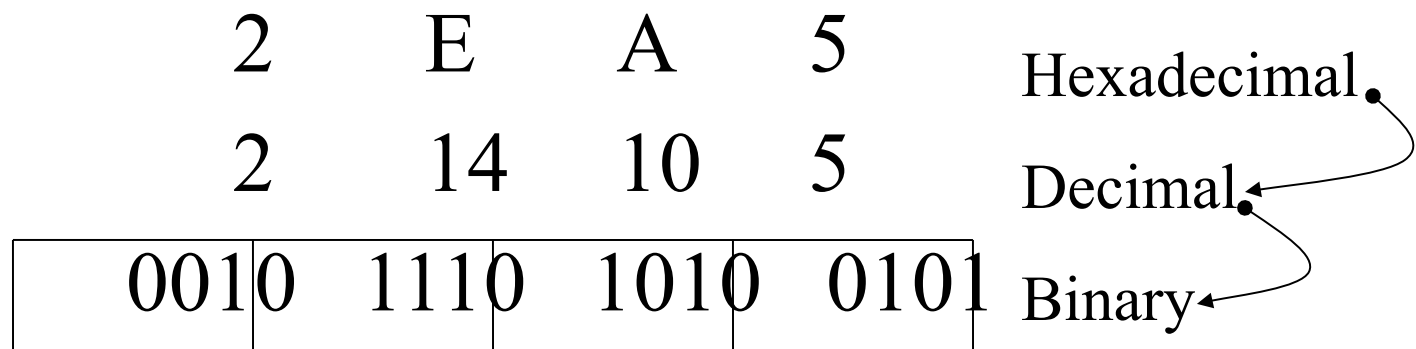
Example



Conversion between number systems – 6

- From hexadecimal to binary
 - Convert each digit in hex to decimal;
 - Convert each decimal to binary with 4-bits;
 - Order the groups from **the least significant bit**.

Example



Conversion between number systems – 7

- From octal to binary
 - Convert each digit in octal to binary with 3-bits;
 - Order the groups from the least significant bit.

Example: convert 2735_8 to binary

				Octal
2	7	3	5	Binary
				
	010	111	011	101

The binary result is 010111011101.

Internal representation of character

Internal representation of character – ASCII 7-bit code

- ASCII: American Standard Code for Information Interchange;
- 128 characters in the code;
- A unique 7-bit binary code for each character which is stored in a byte (8-bits);
- All can be generated from a keyboard and some can be interpreted by monitors and printers;
- Non printable characters, *e.g.* return, tab, spacebar...;
- Letter and number sequences of ASCII are ordered.

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Internal representation of characters

– other codes

- First attempt at increasing the number of printable characters was ISO 8859-1, which supports most western European languages.
- Further attempts increase the coverage to include eastern European languages, Cyrillic, Arabic and Hebrew.
- Unicode (ISO/IEC 10646) *e.g.* UTF-8, UTF-16, etc. covers all the world's major written languages. It is still developing.
- Think about: how to encode different fonts and styles of English characters?

End of the lecture

Questions and exercises – 1

- Convert the following decimal numbers to hexadecimal and binary numbers:
(a) 39; (b) 59; (c) 512; (d) 122; (e) 337; (f) 678
- Convert the following binary numbers to hexadecimal and decimal numbers:
(a) 00011011 (b) 00010111 (c) 10010001 (d) 11100110
(e) 0110 0010 1111 0000 (f) 1010 0011 1100 1110
- Convert the following octal numbers (8-base numbers) to hexadecimal:
(a) 3653_8 (b) 1766_8 (c) 2463_8

End of the lecture

Questions and exercises – 2

- The follow code is a message in ASCII represented in the hexadecimal notation. What does it say?

49 20 61 6D 20 61 20 43 53 20 73 74 75 64 65 6E 74 2E

- Encode the following sentences in ASCII using one byte per character.

Is 1 byte = 8 bits?

Yes, a byte contains 8 bits.